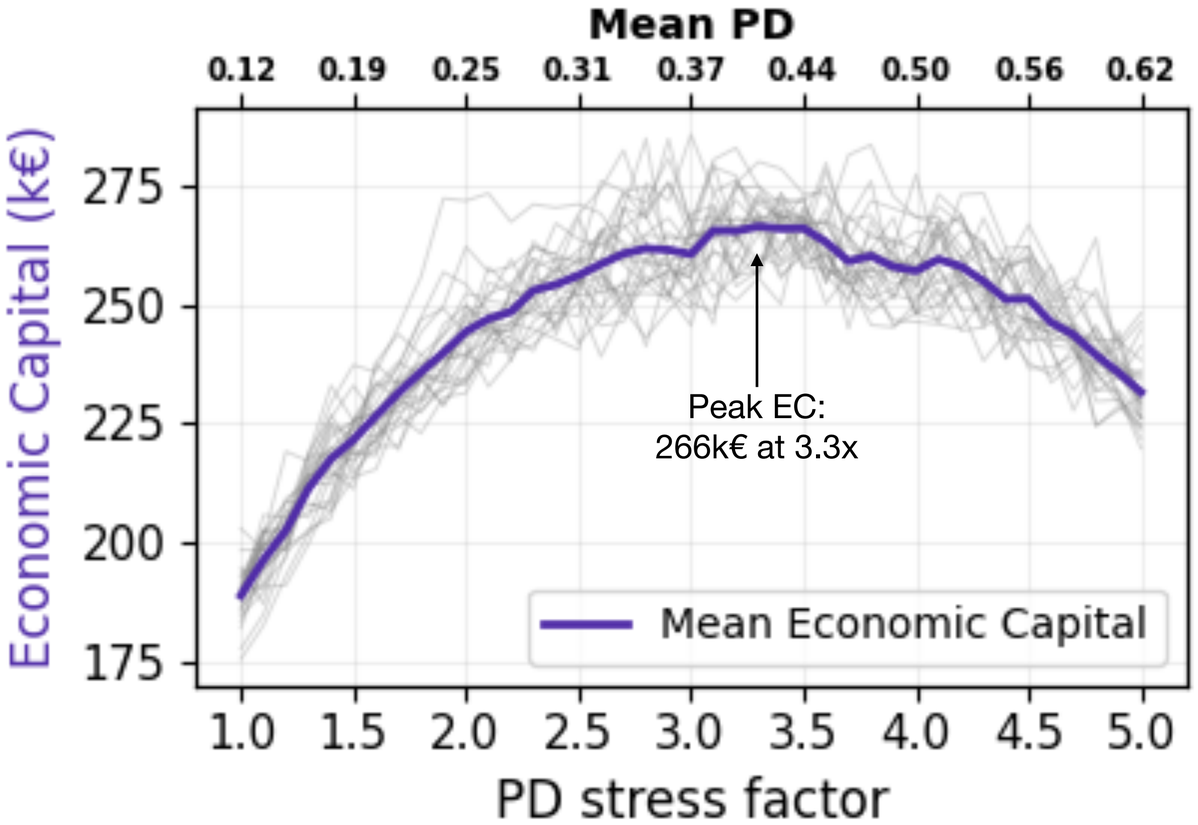
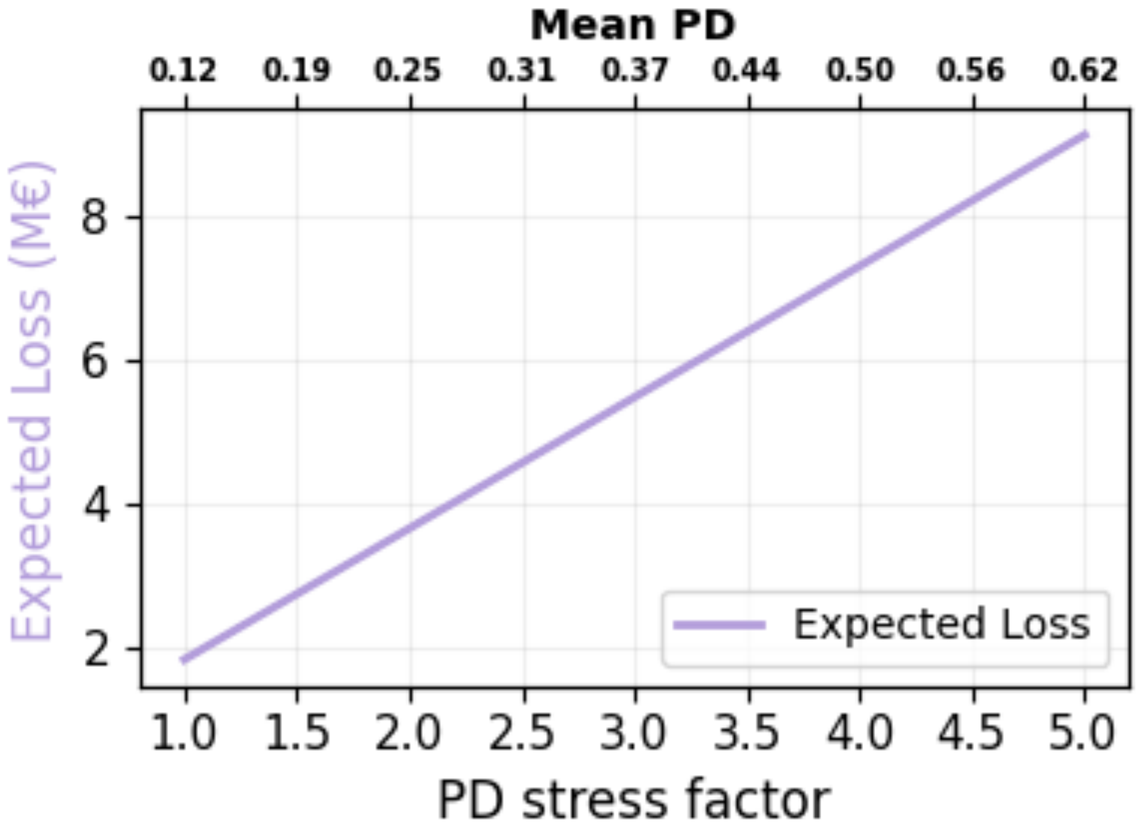
Monte Carlo-Based PD Stress Testing

Economic Capital Response to Increasing PD Stress

	Economic Capital	Expected Loss
baseline	189 k€	1,83 M€
maximum increase	+41 %	+400 %
stress factor at reported maximum	3.3 x	5.0 x

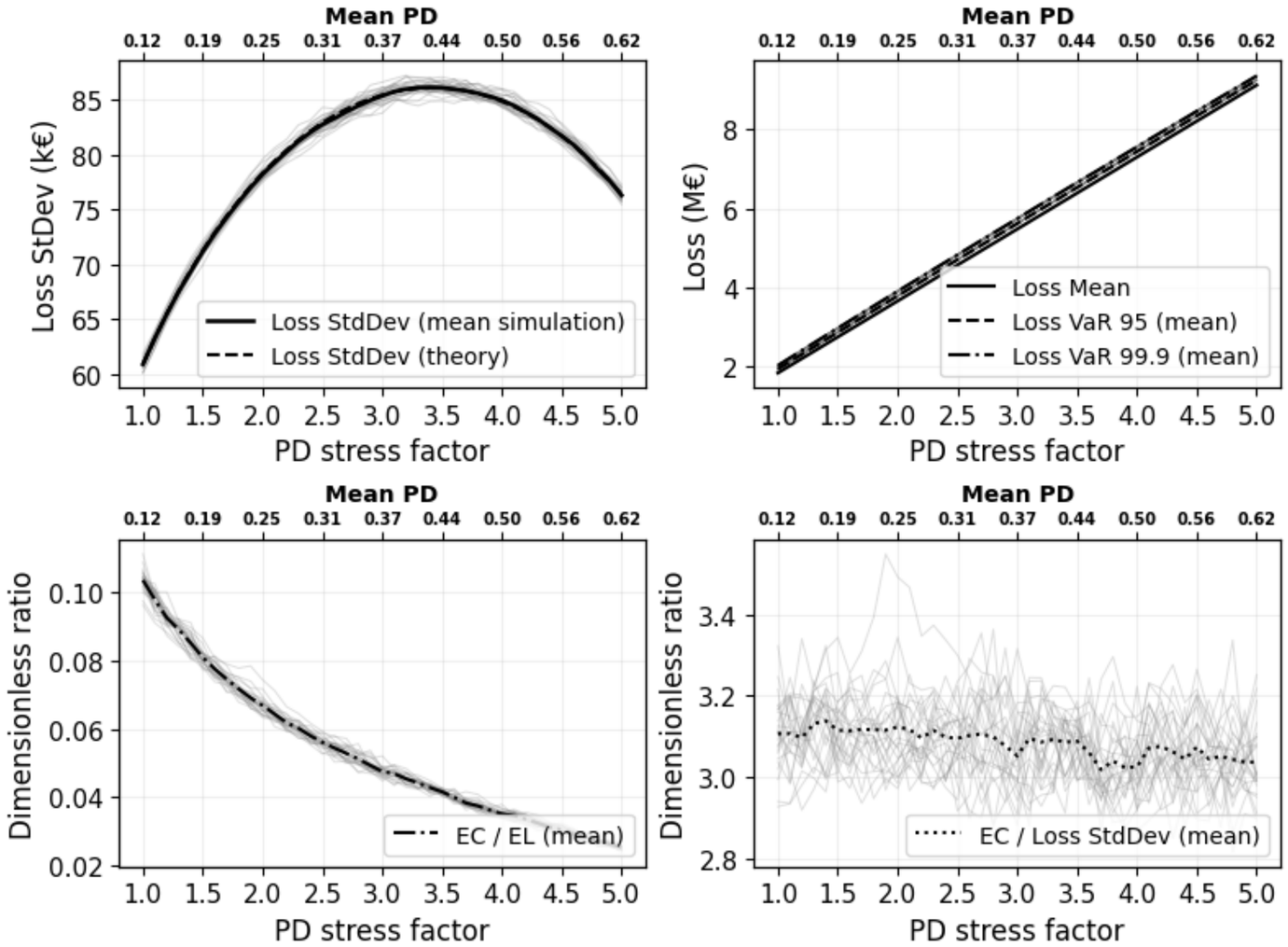




Expected Loss increases monotonically, while Economic Capital peaks and subsequently declines under increasing PD stress.

Monte Carlo-Based PD Stress Testing

Simulation Setup and Validation



Economic Capital Definition:

$$EC = VaR_{99.9\%} - EL$$

Theoretical Loss Standard Deviation:

$$\sigma_L = \sqrt{\sum_{i=1}^{N} (EAD_i \cdot LGD_i)^2 \cdot PD_i(1 - PD_i)}$$

Portfolio	approved heterogeneous
PD threshold	20 %
PD stress factors	1.0-5.0 x
Stress scenarios	41
Seeds	25
Simulations	10,000 per scenario/seed*
Default model	independent Bernoulli

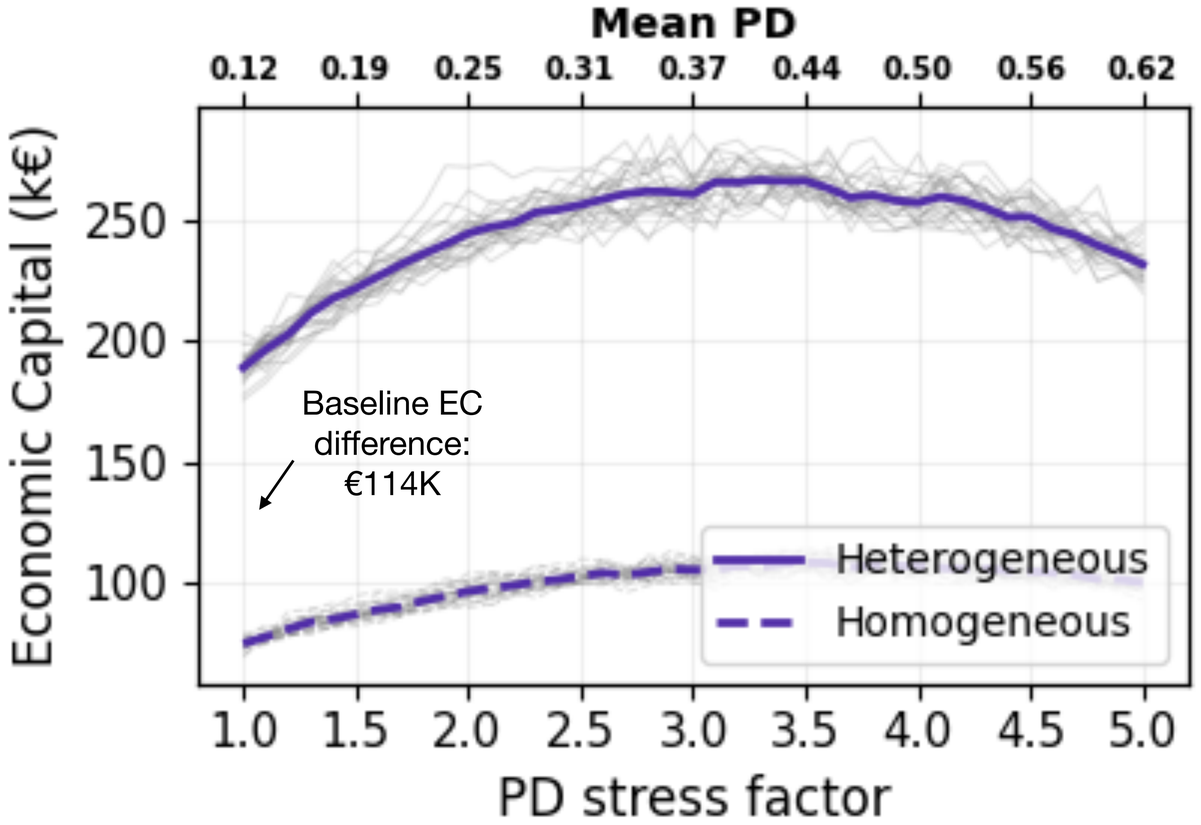
Simulated loss standard deviation closely matches its theoretical counterpart under the independent-default assumption.
*VaR 99.9% estimates are subject to finite-simulation tail uncertainty.

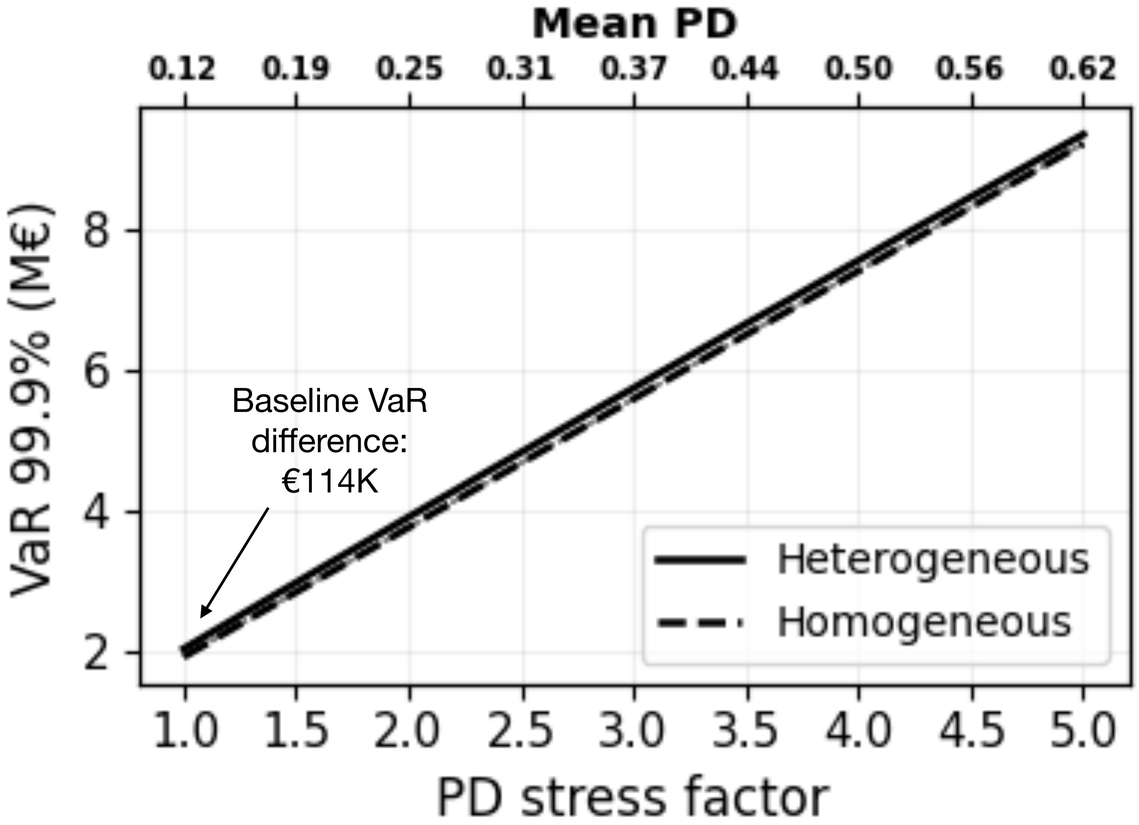
Konstantinos Kazanas

Monte Carlo-Based PD Stress Testing

Sensitivity to Portfolio EAD and LGD Heterogeneity

	Heterogeneous	Homogeneous
baseline	189 k€	75 k€
maximum increase	+41 %	+44 %
stress factor at reported maximum	3.3 x	3.5 x





Preserving Expected Loss does not preserve Economic Capital. Homogenising EAD and LGD produces a lower VaR, resulting in substantially lower Economic Capital despite an unchanged Expected Loss.